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Name: _____

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Department/School: _____

SID: _____

University Physics

Final Examination

Kuang Yaming Honors School, Nanjing University

Select five out of following six problems.

1.(20 pts.) A rod of length L_0 moves with a speed v along the x -direction. The rod makes an angle θ_0 with the x' -axis.

(a) Show that the length of the rod as measured by a stationary observer is given by

$$L = L_0 \left[1 - (v^2/c^2) \cos^2 \theta_0 \right]^{1/2}.$$

(b) Show that the angle that the rod makes with the x -axis is $\theta = \tan^{-1}(\gamma \tan \theta_0)$ where $\gamma = 1/\sqrt{1-v^2/c^2}$.

Solution: (a) Denoting the coordinates of two endpoints of the rod in the S' frame as (x'_1, y'_1) and (x'_2, y'_2) , we have

$$\Delta x' = x'_2 - x'_1 = L_0 \cos \theta_0, \quad \Delta y' = y'_2 - y'_1 = L_0 \sin \theta_0.$$

Using the Lorentz transformation

$$\Delta x' = \gamma(\Delta x - v\Delta t), \quad \Delta y' = \Delta y$$

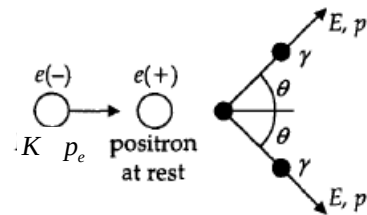
and putting $\Delta t = 0$ (simultaneous determination of the endpoint coordinates in the S frame), we have

$$\begin{aligned} L = [(\Delta x)^2 + (\Delta y)^2]^{1/2} &= [(\Delta x'/\gamma)^2 + (\Delta y')^2]^{1/2} = [L_0^2 \cos^2 \theta_0 (1 - v^2/c^2) + L_0^2 \sin^2 \theta_0]^{1/2} \\ &= L_0 [1 - (v^2/c^2) \cos^2 \theta_0]^{1/2}. \end{aligned}$$

$$(b) \quad \tan \theta = \frac{\Delta y}{\Delta x} = \frac{\Delta y'}{\Delta x'/\gamma} = \gamma \frac{L_0 \sin \theta_0}{L_0 \cos \theta_0} = \gamma \tan \theta_0, \text{ and then}$$

$$\theta = \tan^{-1}(\gamma \tan \theta_0).$$

2.(20 pts.) An electron having kinetic energy $K = 1.000 \text{ MeV}$ makes a head on collision with a positron at rest. In the collision the two particles annihilate each other into two γ -ray photons of equal energy, each traveling at equal angles with the electron's direction of motion, as shown in the figure. Find the energy E , momentum p and angle of emission of the γ -rays. (The electron rest mass is $0.511 \text{ MeV}/c^2$.)



Solution: Conservation of mass energy requires

$$K + 2m_e c^2 = 2E.$$

Thus, the energy of the γ ray is

$$E = \frac{K}{2} + m_e c^2 = 0.500 \text{ MeV} + 0.511 \text{ MeV} = 1.011 \text{ MeV}.$$

Using the energy-momentum relation for the photon, we have

$$p = \frac{E}{c} = 1.011 \text{ MeV}/c.$$

Conservation of momentum requires $p_e = 2p \cos \theta$ where p_e is the initial momentum of the electron. Using the energy momentum relation for the electron

$$E_e^2 = m_e^2 c^4 + c^2 p_e^2$$

where E_e is the electron total energy $E_e = K + m_e c^2$, we have

$$p_e = \frac{1}{c} \sqrt{K^2 + 2Km_e c^2} = \frac{1}{c} \sqrt{(1.000)^2 + 2(1.000)(0.511)\text{MeV}} = 1.422\text{MeV}/c.$$

Therefore,

$$\cos \theta = \frac{p_e}{2p} = 0.703, \text{ or } \theta = 45.3^\circ.$$

3.(20 pts.) A mole of the ideal monatomic gas occupies a volume V_i at a pressure p_i . The gas undergoes a process in which the pressure is proportional to the volume. At the end of the process, it is found that the root-mean-square speed v_{rms} of the gas molecules has doubled from its initial value. Determine the amount of energy transferred to the gas by heat.

Solution: Since

$$v_{\text{rms}} = \sqrt{\frac{3k_B T}{m}},$$

the temperature increases by factor 4 as the rms speed doubles: $T_f = 4T_i$. Since the pressure is proportional to the volume in the process,

$$p = \frac{p_i}{V_i} V.$$

Using the equation of state, $pV = \frac{p_i}{V_i} V^2 = nRT$, we have $\frac{V_f^2}{V_i^2} = \frac{T_f}{T_i} = 4$ and then

$V_f = 2V_i$. The work done by the gas is

$$W = \int_{V_i}^{V_f} p dV = \frac{p_i}{V_i} \int_{V_i}^{V_f} V dV = \frac{p_i}{2V_i} [(2V_i)^2 - V_i^2] = \frac{3p_i V_i}{2}.$$

The change of internal energy of the system is

$$\Delta U = C_v (T_f - T_i) = \frac{3R}{2} (3T_i) = \frac{9RT_i}{2} = \frac{9p_i V_i}{2}.$$

Thus, the amount of energy transferred to the gas by heat is

$$Q = \Delta U + W = 6p_i V_i.$$

4.(20 pts.) The process H_2O (liquid) (-10°C) $\rightarrow$ H_2O (solid) (-10°C) of 1 mol of H_2O is carried out in an isolated system consisting of the 1 mol of H_2O and a thermostat at constant temperature of -10°C . The latent heat for melting ice is 333J/g . The specific heat capacity of water is $4.2\text{J/g}^\circ\text{C}$ and the specific heat capacity of ice is $2.1\text{J/g}^\circ\text{C}$.

(a) Calculate the change in entropy of the 1 mol of H_2O in this process.

(b) Calculate the change in entropy of the thermostat.

(c) Is this process spontaneous? Why?

(Hint: The process of freezing of the super cooled water is irreversible. Thus, to calculate the entropy change in the process H_2O (liquid) (-10°C)(state 1) $\rightarrow$ H_2O (solid) (-10°C)(state 2), one has to find a reversible path connecting the state 1 and state 2.)

Solution: (a) The a reversible path connecting the state 1 and state 2 is given below:

Step 1 : liquid at $-10^\circ\text{C} \rightarrow$ liquid at 0°C , $Q_1 = c_{\text{water}}m(T_2 - T_1)$

Step 2: liquid at $0^\circ\text{C} \rightarrow$ solid at 0°C , $Q_2 = -Lm$

Step3: solid at $0^\circ\text{C} \rightarrow$ solid at -10°C $Q_3 = c_{\text{ice}}m(T_1 - T_2)$

Thus, the change in entropy of the 1 mol of H_2O is

$$\Delta S = c_{\text{water}}m \ln\left(\frac{T_2}{T_1}\right) - \frac{Lm}{T_2} + c_{\text{ice}}m \ln\left(\frac{T_1}{T_2}\right) = -20.55 \text{ J/K}.$$

(b) The change in entropy of the thermostat is

$$\Delta S_{\text{therm}} = \frac{-(Q_1 + Q_2 + Q_3)}{T_1} = 21.35 \text{ J/K}.$$

(c) The total entropy change of the isolated system(H_2O + thermostat) is

$$\Delta S_{\text{total}} = \Delta S + \Delta S_{\text{therm}}.$$

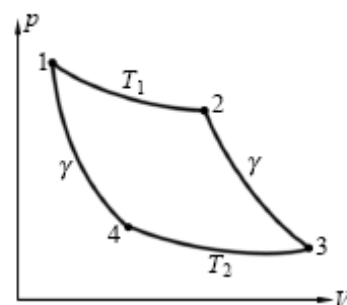
Since it is positive, the process is spontaneous.

5.(20 pts.) Assume that the internal energy of a mole of gas that obeys the Van der Waals equation of state

$$\left(p + \frac{a}{V^2}\right)(V - b) = RT$$

is given by

$$U = U_0 + C_V T - \frac{a}{V}$$



where C_V is considered to be a constant. The gas undergoes a Carnot cycle between two heat reservoirs of temperature T_1 and T_2 , as shown in the accompanying figure.

(a) Show that the adiabatic equation of the gas is given by $T(V - b)^{R/C_V} = C$.

(b) Calculate the work done by the system and the heat input for the cycle.

(c) Obtain the thermal efficiency for the cycle in terms of T_1 and T_2

Solution: (a) From

$$U = U_0 + C_V T - \frac{a}{V},$$

we have

$$dU = C_V dT + \frac{a}{V^2} dV.$$

Using the first law, we have

$$dQ = dU + pdV = C_v dT + \frac{a}{V^2} dV + \left(\frac{RT}{V-b} - \frac{a}{V^2} \right) dV = C_v dT + \frac{RT}{V-b} dV$$

For a adiabatic process, $dQ = 0$. Then

$$C_v dT + \frac{RT}{V-b} dV = 0 \quad \text{or} \quad \frac{dT}{T} + \frac{R}{C_v} \frac{dV}{V-b} = 0.$$

Thus $T(V-b)^{\frac{R}{C_v}} = C$.

(b) $1 \rightarrow 2$: $\Delta Q_{12} = \Delta W_{12} + (U_2 - U_1)$,

$$\Delta W_{12} = \int_{V_1}^{V_2} p dV = \int_{V_1}^{V_2} \left(\frac{RT_1}{V-b} - \frac{a}{V^2} \right) dV = RT_1 \ln \frac{V_2-b}{V_1-b} + \left(\frac{a}{V_2} - \frac{a}{V_1} \right),$$

$$U_2 - U_1 = -\frac{a}{V_2} + \frac{a}{V_1}, \quad \Delta Q_{12} = RT_1 \ln \frac{V_2-b}{V_1-b}.$$

$3 \rightarrow 4$: $\Delta Q_{34} = \Delta W_{34} + (U_4 - U_3) = RT_2 \ln \frac{V_4-b}{V_3-b}$.

For a complete cycle, $\Delta U = 0$. By using the first law, we have

$$W = \Delta Q_{12} + \Delta Q_{34} = RT_1 \ln \frac{V_2-b}{V_1-b} - RT_2 \ln \frac{V_3-b}{V_4-b}.$$

The heat input for one cycle is $Q_{in} = \Delta Q_{12} = RT_1 \ln \frac{V_2-b}{V_1-b}$.

(c) The efficiency is $\eta = \frac{W}{Q_{in}} = \frac{T_1 \ln \frac{V_2-b}{V_1-b} - T_2 \ln \frac{V_3-b}{V_4-b}}{T_1 \ln \frac{V_2-b}{V_1-b}}$.

Using the adiabatic equation, we have

$$\begin{aligned} T_1(V_2-b)^{\frac{R}{C_v}} &= T_2(V_3-b)^{\frac{R}{C_v}} \\ T_1(V_1-b)^{\frac{R}{C_v}} &= T_2(V_4-b)^{\frac{R}{C_v}} \end{aligned}$$

Then we have

$$\frac{V_2-b}{V_1-b} = \frac{V_3-b}{V_4-b}.$$

Therefore, we have

$$\eta = \frac{W}{Q_{in}} = \frac{T_1 - T_2}{T_1} = 1 - \frac{T_2}{T_1}.$$

6.(20 pts.) A mole of gas obeys equations

$$\left(\frac{\partial V}{\partial T}\right)_p = \frac{R}{p} + \frac{a}{T^2}, \quad \left(\frac{\partial V}{\partial p}\right)_T = -Tf(p)$$

where a is a constant and $f(p)$ is a function of pressure p only.

(a) Find $f(p)$.

(b) Find the equation of state of the system.

(c) The constant-pressure (isobaric) heat capacity of the system can be expressed as

$$C_p = T \left(\frac{\partial S}{\partial T}\right)_p. \text{ Show that } \left(\frac{\partial C_p}{\partial p}\right)_T = \frac{2a}{T^2}.$$

Solution: (a) Using the relation $\left[\frac{\partial}{\partial T} \left(\frac{\partial V}{\partial p}\right)_T\right]_p = \left[\frac{\partial}{\partial p} \left(\frac{\partial V}{\partial T}\right)_p\right]_T$, we have

$$-f(p) = \left[\frac{\partial}{\partial p} \left(\frac{R}{p} + \frac{a}{T^2}\right)\right]_T = -\frac{R}{p^2} \quad \text{or} \quad f(p) = \frac{R}{p^2}.$$

(b)

$$\begin{aligned} dV &= \left(\frac{\partial V}{\partial T}\right)_p dT + \left(\frac{\partial V}{\partial p}\right)_T dp = \left(\frac{R}{p} + \frac{a}{T^2}\right) dT - \frac{RT}{p^2} dp \\ &= \frac{R}{p} dT - d\left(\frac{a}{T}\right) + Td\left(\frac{R}{p}\right) = d\left(\frac{RT}{p} - \frac{a}{T}\right). \end{aligned}$$

Thus

$$V = \frac{RT}{p} - \frac{a}{T} + C.$$

Since $T \rightarrow \infty$, $V \rightarrow \frac{RT}{p}$ (ideal gas), we have $C = 0$. Therefore, the equation of state is

$$V = \frac{RT}{p} - \frac{a}{T}.$$

$$(c) \quad \left(\frac{\partial C_p}{\partial p}\right)_T = \left[\frac{\partial}{\partial p} \left(T \left(\frac{\partial S}{\partial T}\right)_p\right)\right]_T = T \left[\frac{\partial}{\partial p} \left(\frac{\partial S}{\partial T}\right)_p\right]_T = T \left[\frac{\partial}{\partial T} \left(\frac{\partial S}{\partial p}\right)_T\right]_p.$$

Using the Maxwell relation $\left(\frac{\partial S}{\partial p}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_p$, ($dG = -SdT + Vdp$)

$$\left(\frac{\partial C_p}{\partial p}\right)_T = -T \left[\frac{\partial}{\partial T} \left(\frac{\partial V}{\partial T}\right)_p\right]_p = -T \left[\frac{\partial}{\partial T} \left(\frac{R}{p} + \frac{a}{T^2}\right)\right]_p = -T \left(-2\frac{a}{T^3}\right) = \frac{2a}{T^2}.$$